

KITEWRIGHT CORE · BUILD PLAN

Kitewright Core prototype build plan

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	First build plan; design made constructable (KWC-DDR-002)

Kitewright Core prototype build plan

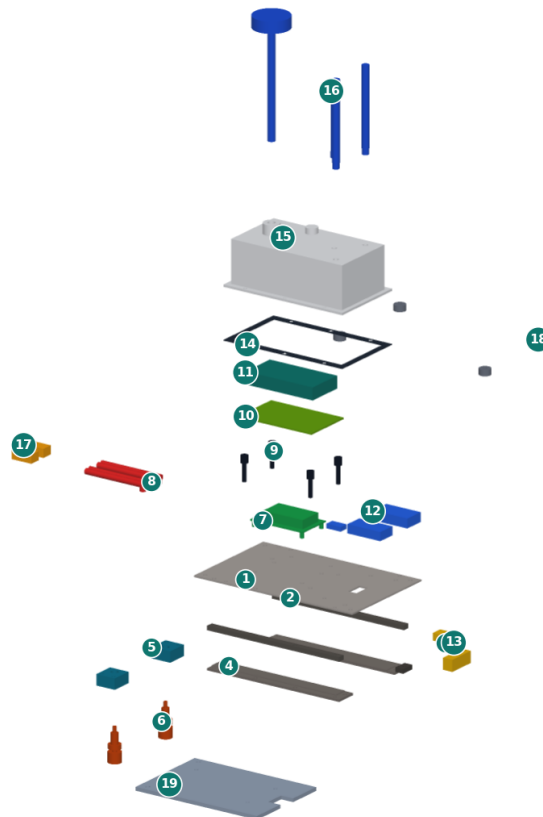
Plan, not yet built. How to build the first proof-of-concept prototype, component by component. Building and testing to it is TRL 4 work. Decisions still to be made are kept in the design decisions register ([docs/06-design-decisions.md](#)), not here.

■ 1. What you are building

Kitewright Core prototype: every component, pulled apart

Numbered in build order. Seen from the front right and above

- 1 ■ Core plate
- 2 ■ Rail spacer bars (2)
- 3 ■ Front stops (2)
- 4 ■ Rail lips (2) with wear tape
- 5 ■ Pin blocks (2)
- 6 ■ Locking pins (2) and jam nuts
- 7 ■ Power board on 8 mm standoffs
- 8 ■ Power leads (4)
- 9 ■ 25 mm standoffs and dampers (4)
- 10 ■ Damping plate
- 11 ■ Flight controller
- 12 ■ Radio, power modules, receiver
- 13 ■ DS-014 pigtail and plug
- 14 ■ Lid gasket
- 15 ■ Lid with grommets and safety switch
- 16 ■ Antennas, GNSS mast and receiver
- 17 ■ AS150 plugs (4)
- 18 ■ Corner spacers (4)
- 19 ■ Payload shoe



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github.com/BoujeeEnjinia1701/kitewright-core

Figure 1. Every component of the core, pulled apart and numbered in build order.

The prototype is one Kitewright Core: a 240 x 150 mm aluminium plate with the avionics on top under a printed lid, and the payload rail and two locking pins underneath, plus one payload shoe. Figure 1 shows its 19 components in the order you make or fit them. Eight are made in a small workshop: the core plate, the rail spacer bars, front stops, rail lips, pin blocks, payload shoe, lid and damping plate. Everything else is bought and fitted: the flight controller, GNSS receiver, power board and modules, radios, antennas, the DS-014 pigtail, the locking pins (indexing plungers), dampers, standoffs, clinch nuts, leads and connectors. The work is sawing, drilling, tapping and filing aluminium

sheet and bar, pressing clinch nuts, one 3D print, cutting a glass-epoxy plate, and soldering heavy power leads. The parts cost about USD 1,760 for the airborne core from the bill of materials; the ground station kit is bought complete.

Safety: The finished core switches lithium pack power of up to 60 V and 200 A. Until section 6 says otherwise, power it only from a current-limited bench supply, never from a pack. Soldering 8 AWG leads needs a 100 W iron or better and gets the wire hot enough to burn: hold it with pliers. Cut aluminium edges and glass-epoxy dust are hazards: deburr every edge, wear gloves and a dust mask, and cut G10 wet or under extraction. Printing ASA gives off fumes; print in a ventilated space.

■ 2. What changed to make it buildable

The scaffold described the core in words; to build it, each part needed a shape, a material and a fixing. Each change below keeps what the core does and is recorded in decision record KWC-DDR-002.

Table 1. Changes from the concept.

Component	The concept had	The buildable design has	Why
Payload rail	"A plain rail"	Spacer bar and taped lip on each side, bolted under the plate; 0.3 mm gap above the shoe (Figure 11)	Made from stock bar; the gap comes from stock thicknesses
Locking pin	"A locking pin"	Two spring plungers in blocks under the lips (Figure 13)	Either pin alone holds the payload
End of travel	None	Two front stops between plate and lip (Figure 14)	Lines the pins up with their holes
Payload connector	A connector on the mount	A pigtail through a slot in the plate and a notch in the shoe (Figure 20)	A sliding shoe cannot mate a 40-pin plug reliably
Fixings under the plate	Not shown	Clinch nuts pressed in flush from below (Figure 22)	The shoe slides just under the plate
Core to frame	"Bolts into any frame"	Hangs under the deck on four bolts and spacers (Figure 27)	Payload load goes straight into the frame
Flight controller	Not shown	On four dampers and a glass-epoxy plate above the power board (Figure 17)	Vibration isolation
Lid	"Cold-rated enclosure"	Printed lid with flange, gasket and six screws	Clears the stop nuts and the deck opening

■ 3. Making the components

Sizes are in millimetres. Positions are given from the centre of the core plate: "along" means toward the front (the end with the cable slot), "across" means to the left or right of the centre line.

3.1 Core plate

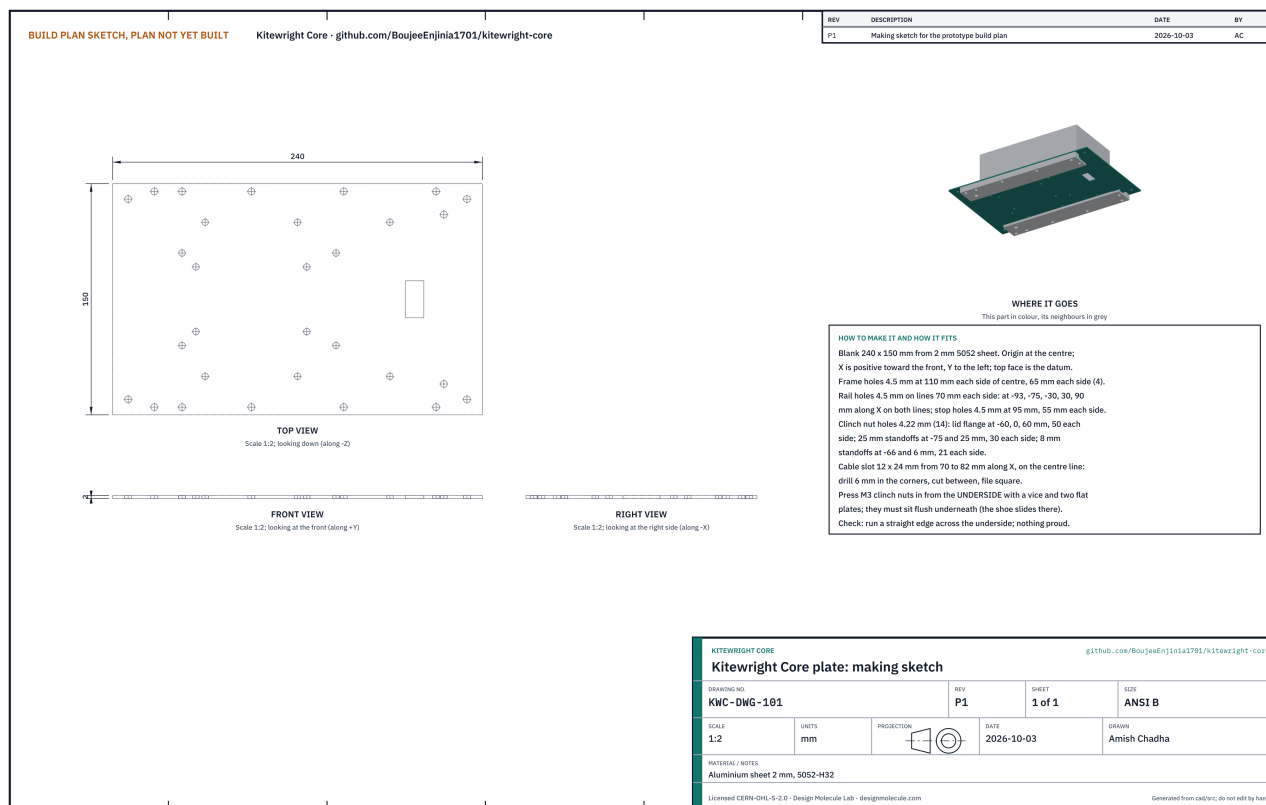


Figure 2. Making sketch KWC-DWG-101.

What it is and what it is made from. The flat plate everything fixes to: 240 x 150 mm, cut from 2 mm 5052-H32 aluminium sheet.

How to make it.

1. Cut the blank 240 x 150 mm and mark a centre line each way.
2. Drill the four frame holes 4.5 mm, 110 mm along each way from the centre and 65 mm across each way.
3. Drill the rail holes 4.5 mm on two lines 70 mm each side of the centre line, at 93 mm and 75 mm behind the centre, 30 mm behind, 30 mm ahead and 90 mm ahead. Drill the two stop holes 4.5 mm at 95 mm ahead, 55 mm each side.
4. Drill the 14 clinch nut holes 4.22 mm: six for the lid flange (60 mm behind, on the centre, 60 mm ahead; 50 mm each side), four for the tall standoffs (75 mm behind and 25 mm ahead; 30 mm each side) and four for the short standoffs (66 mm behind and 6 mm ahead; 21 mm each side).
5. Cut the cable slot, 12 mm long and 24 mm wide, from 70 to 82 mm ahead of the centre: drill 6 mm holes in its corners, cut between them and file square.
6. Deburr every hole and edge.
7. Press an M3 clinch nut into each 4.22 mm hole **from the underside**, in a vice between two flat steel plates, until its head is flush.

How it fits the parts next to it. The rails bolt to its underside (Figure 11), the lid and standoffs screw into its clinch nuts from above (Figures 17 and 22), and it hangs from the frame on its four corner holes (Figure 27).

Check before moving on. Lay a straight edge across the underside in several directions: nothing stands proud, because the shoe slides 0.3 mm below.

3.2 Rail spacer bars (make 2)

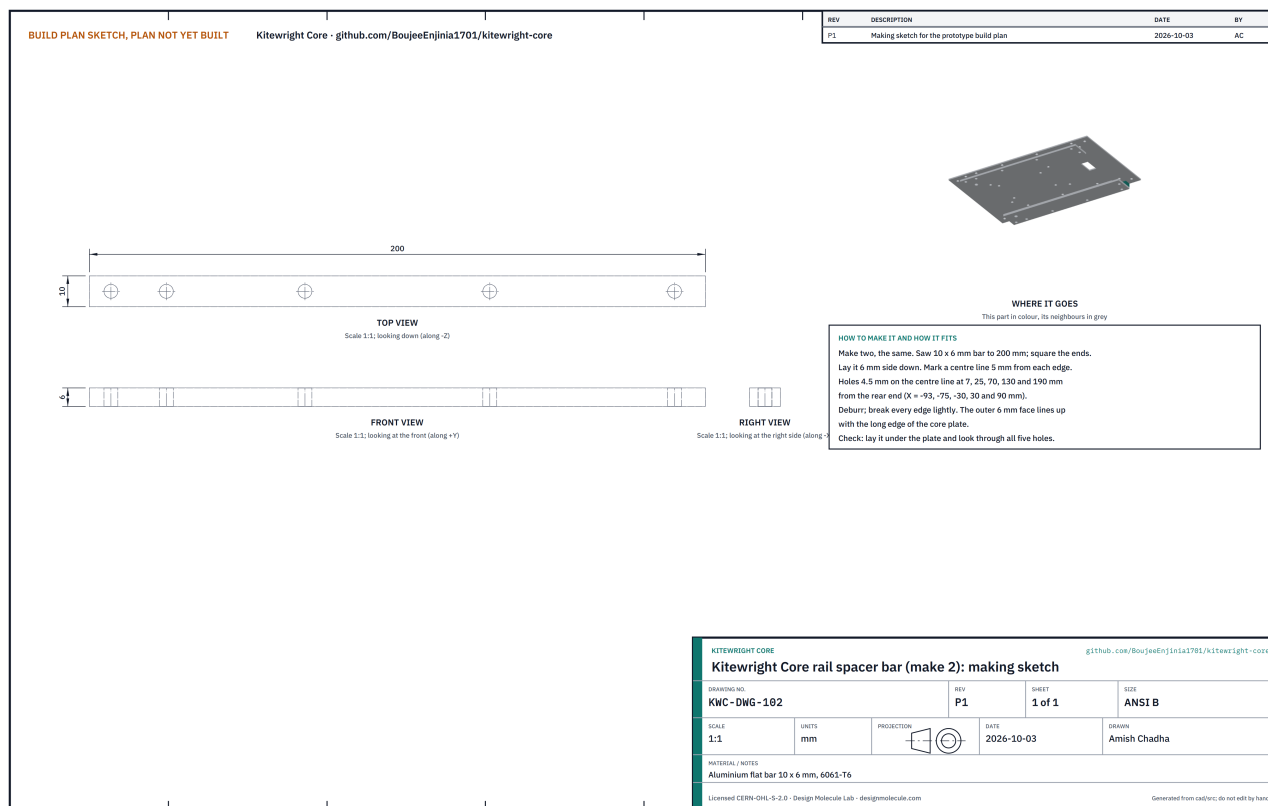


Figure 3. Making sketch KWC-DWG-102.

What it is and what it is made from. The bar that sets the gap between plate and lip: 10 x 6 mm 6061-T6 flat bar, 200 mm long.

How to make it.

1. Saw two pieces 200 mm long and square the ends.
2. Mark a centre line along the 10 mm face and drill five 4.5 mm holes on it at 7, 25, 70, 130 and 190 mm from the rear end.
3. Deburr.

How it fits the parts next to it. Its 10 mm face lies against the underside of the plate with its outer edge on the plate's long edge; the lip lies under it (Figure 11). The screws pass through all three.

Check before moving on. Hold it under the plate and look through all five holes.

3.3 Front stops (make 2)

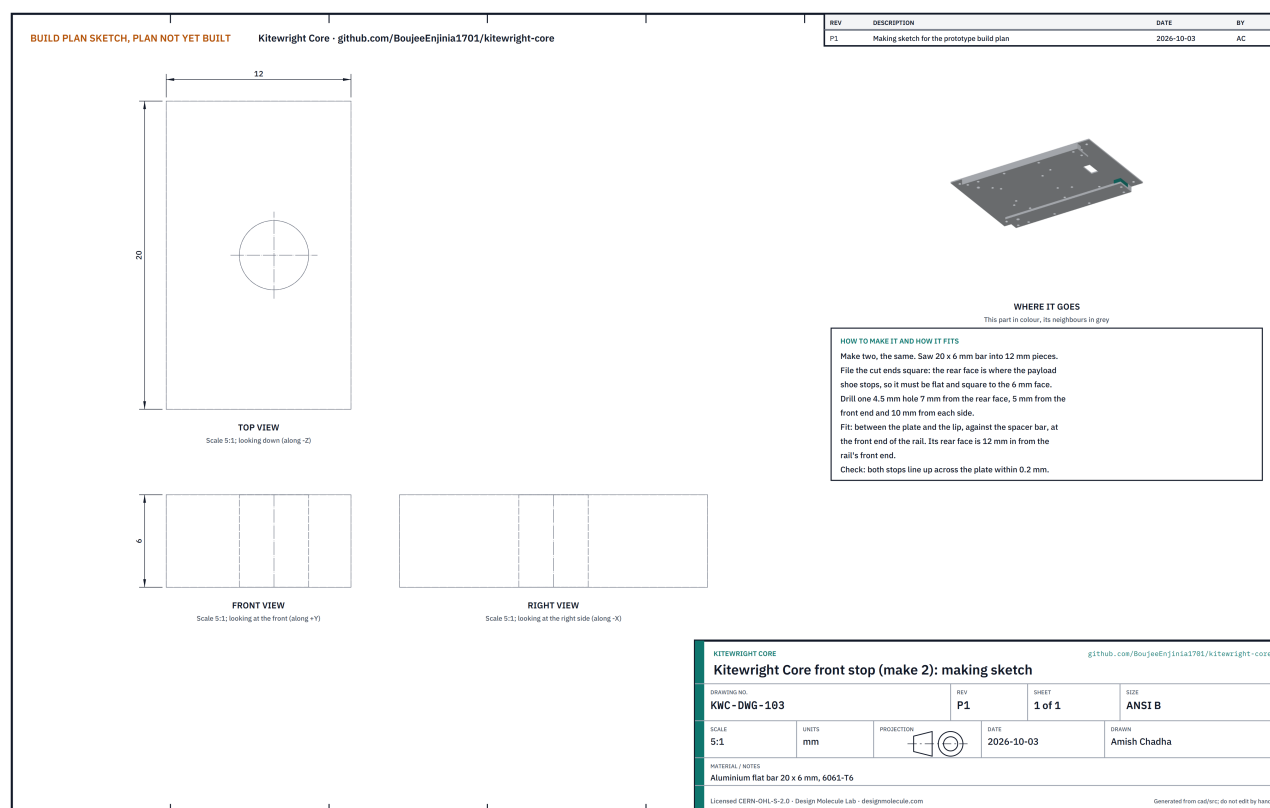


Figure 4. Making sketch KWC-DWG-103.

What it is and what it is made from. A small block that stops the shoe at the front end of the rail: 20 x 6 mm 6061-T6 bar, 12 mm long.

How to make it.

1. Saw two pieces 12 mm long and file the cut ends flat and square: the rear face is where the shoe stops.
2. Drill one 4.5 mm hole, 7 mm from the rear face, 5 mm from the front end and 10 mm from each side.

How it fits the parts next to it. It sits between the plate and the lip, against the inner edge of the spacer bar, with its front end level with the rail's front end. The rail's stop screw clamps it (Figure 14).

Check before moving on. With both stops fitted loosely, their rear faces line up across the plate within 0.2 mm.

3.4 Rail lips with wear tape (make a left and a right)

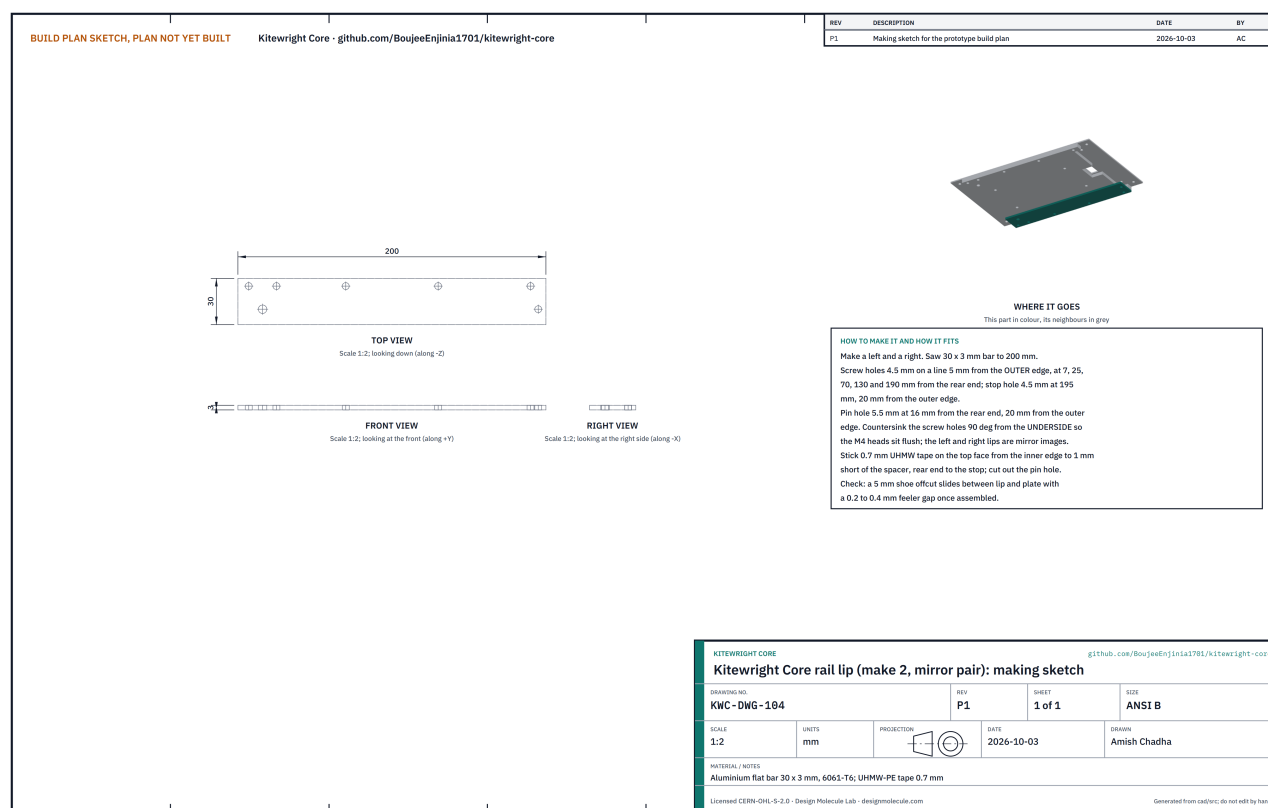


Figure 5. Making sketch KWC-DWG-104.

What it is and what it is made from. The bar the shoe's edge rides on: 30 x 3 mm 6061-T6 flat bar, 200 mm long, with 0.7 mm UHMW-PE tape on top.

How to make it.

1. Saw two pieces 200 mm long.
2. Drill the five screw holes 4.5 mm on a line 5 mm from the outer edge, at 7, 25, 70, 130 and 190 mm from the rear end, and the stop hole 4.5 mm at 195 mm, 20 mm from the outer edge.
3. Drill the pin hole 5.5 mm, 16 mm from the rear end and 20 mm from the outer edge.
4. Countersink the screw and stop holes 90° from the underside, deep enough that an M4 countersunk head sits flush. The left and right lips are mirror images: lay them side by side, outer edges outward, before countersinking.
5. Stick UHMW-PE tape on the top face from the inner edge to 1 mm short of the spacer bar's line, from the rear end to the front stop. Cut out the pin hole with a sharp knife.

How it fits the parts next to it. The tape faces up toward the plate. The shoe's edge overlaps the lip by 19 mm and rides on the tape, with 0.3 mm between the shoe and the plate (Figure 11).

Check before moving on. After step 1 of section 4, a 5 mm offcut of the shoe plate slides between lip and plate along the whole rail, with a 0.2 to 0.4 mm feeler gauge fitting above it.

3.5 Pin blocks (make 2)

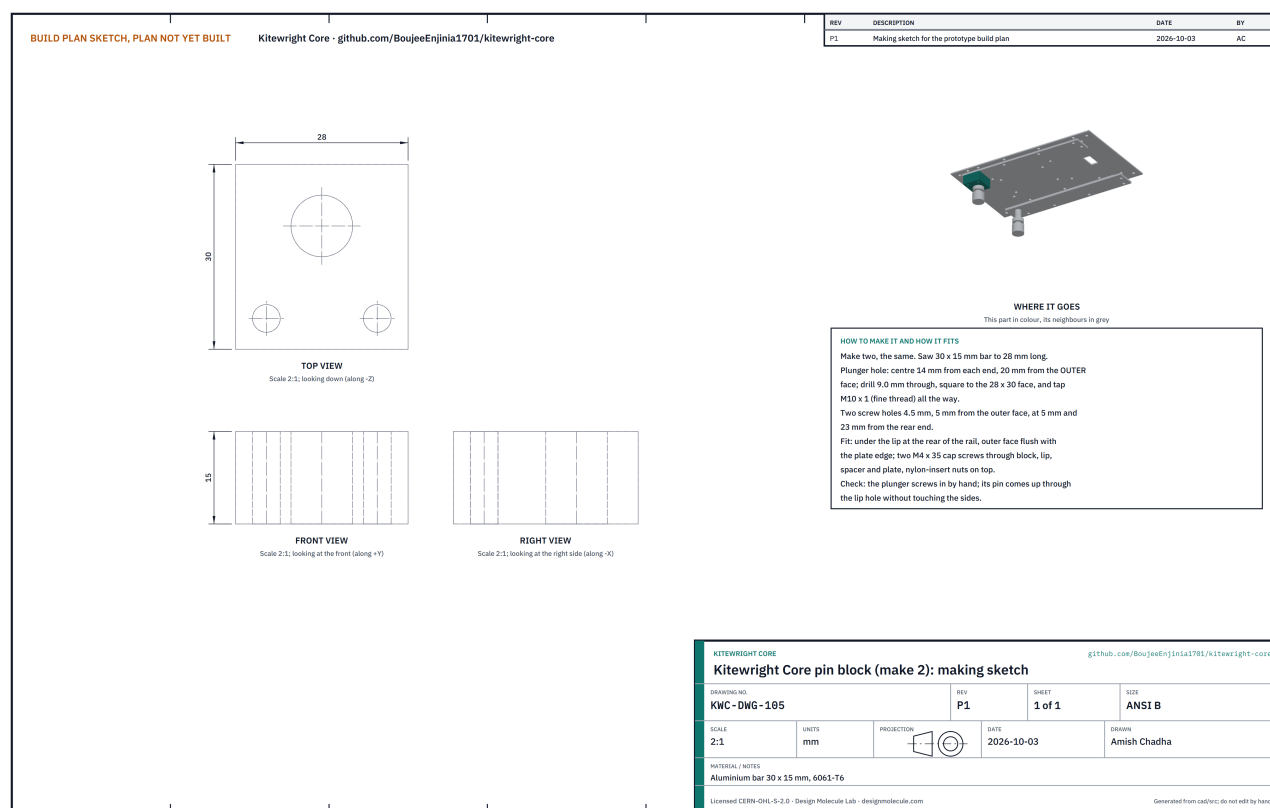


Figure 6. Making sketch KWC-DWG-105.

What it is and what it is made from. The threaded block that holds a locking pin under the lip: 30 x 15 mm 6061-T6 bar, 28 mm long.

How to make it.

1. Saw two pieces 28 mm long.
2. Mark the plunger hole 14 mm from each end and 20 mm from the outer face. Drill it 9.0 mm through, square to the 28 x 30 face, and tap M10 x 1 (fine thread) all the way.
3. Drill two 4.5 mm screw holes 5 mm from the outer face, at 5 and 23 mm from the rear end.

How it fits the parts next to it. The block sits under the lip at the rear of the rail with its outer face flush with the plate's edge; two M4 x 35 cap screws pass up through block, lip, spacer bar and plate into nylon-insert nuts. The plunger screws up into it from below; its pin passes up through the lip and into the shoe (Figure 13).

Check before moving on. The plunger screws in by hand, and its pin comes up through the lip hole without touching the sides.

3.6 Payload shoe

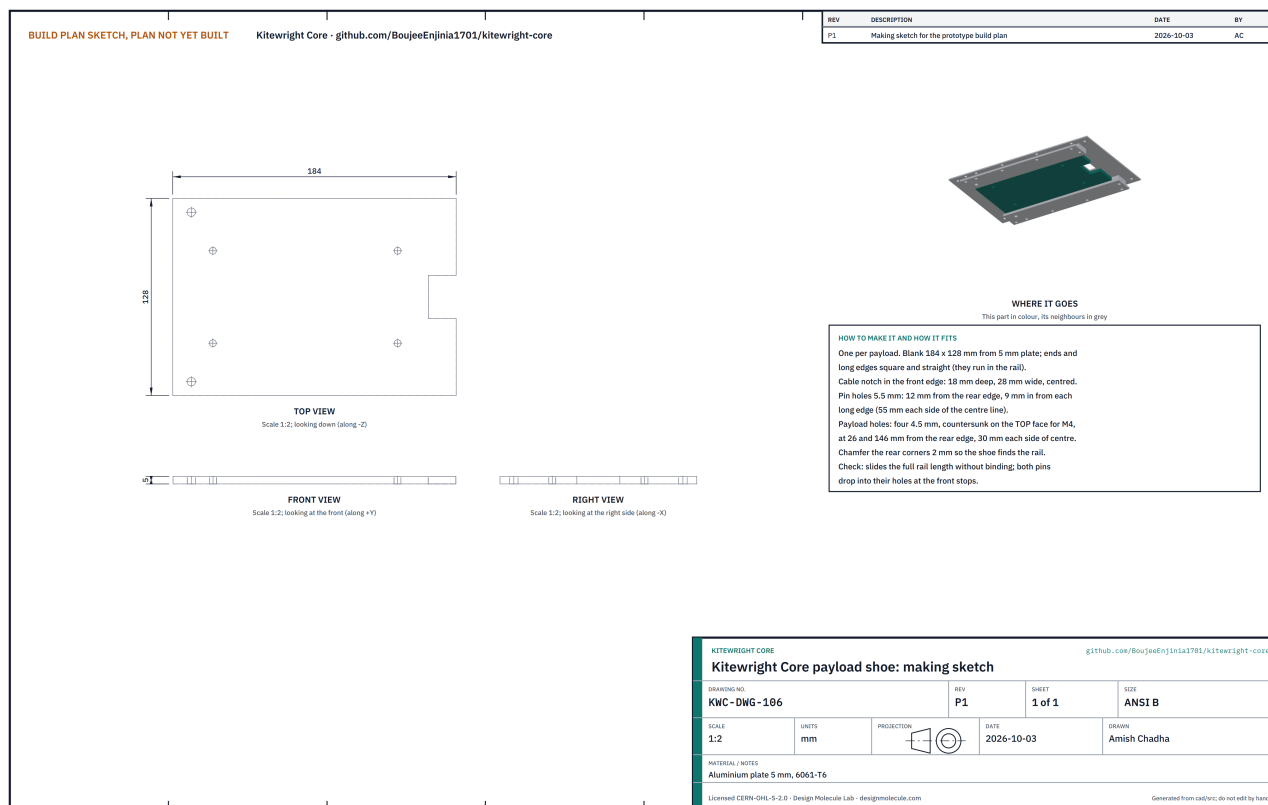


Figure 7. Making sketch KWC-DWG-106.

What it is and what it is made from. The plate every payload carries on its top: 184 x 128 mm, cut from 5 mm 6061-T6 plate. The core kit carries one; each payload maker makes their own to this sketch.

How to make it.

1. Cut the blank 184 x 128 mm. The long edges and the front edge run in the rail and against the stops: make them straight and square.
2. Cut the cable notch in the middle of the front edge, 18 mm deep and 28 mm wide.
3. Drill the two pin holes 5.5 mm, 12 mm from the rear edge and 9 mm in from each long edge.
4. Drill four 4.5 mm payload holes at 26 and 146 mm from the rear edge, 30 mm each side of the centre line, and countersink them from the top so the screw heads sit flush.
5. Chamfer the rear corners 2 mm so the shoe finds the rail.

How it fits the parts next to it. It slides in from the rear between the plate and the lips until its front edge meets the front stops; both pins then rise into its holes (Figures 13 and 14). The payload bolts to its underside with four M4 countersunk screws from the top.

Check before moving on. In section 4, step 10: it slides the full rail without binding, and both pins drop into their holes at the stops.

3.7 Lid

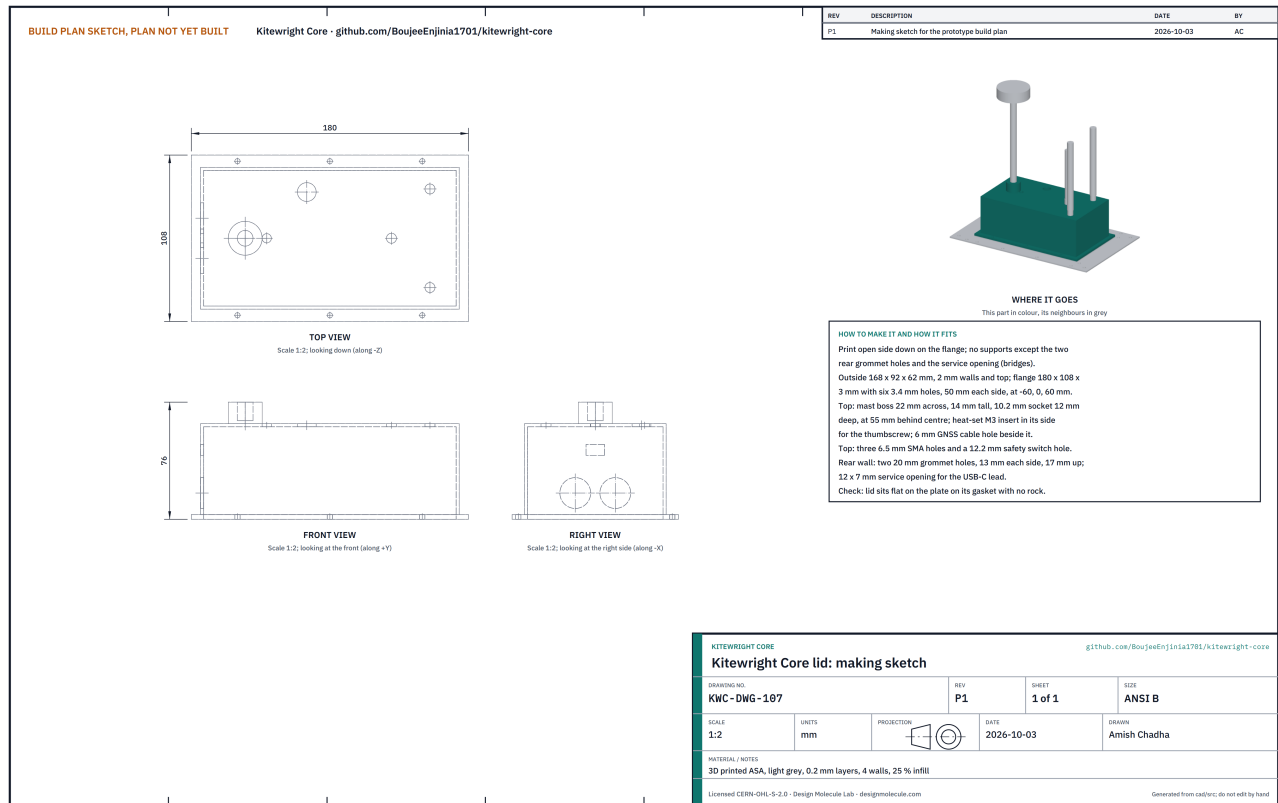


Figure 8. Making sketch KWC-DWG-107.

What it is and what it is made from. The cover over the avionics, printed in light grey ASA: 168 x 92 x 62 mm outside with 2 mm walls and a 180 x 108 x 3 mm flange.

How to make it.

1. Print it open side down on the flange, 0.2 mm layers, four walls, 25 % infill, supports only under the rear grommet holes and the service opening.
2. Check the six 3.4 mm flange holes, the 10.2 mm mast socket (12 mm deep in the 22 mm boss, 55 mm behind the centre), the 6 mm GNSS cable hole beside it, the three 6.5 mm antenna holes and the 12.2 mm switch hole in the top, and the two 20 mm grommet holes (13 mm each side, 17 mm up) and the 12 x 7 mm service opening in the rear wall. Clear them with a drill of the same size.
3. Melt an M3 heat-set insert into the side of the mast boss for the thumbscrew.
4. Cut the 1 mm foam gasket to the flange outline, with six 6 mm holes at the screws.

How it fits the parts next to it. The lid stands on the gasket on the plate and six M3 cap screws pass through its flange into the clinch nuts (Figure 22). It rises through the frame deck's opening.

Check before moving on. The lid sits flat on its gasket with no rock, and the flight controller, once fitted, clears its top by at least 5 mm.

3.8 Damping plate

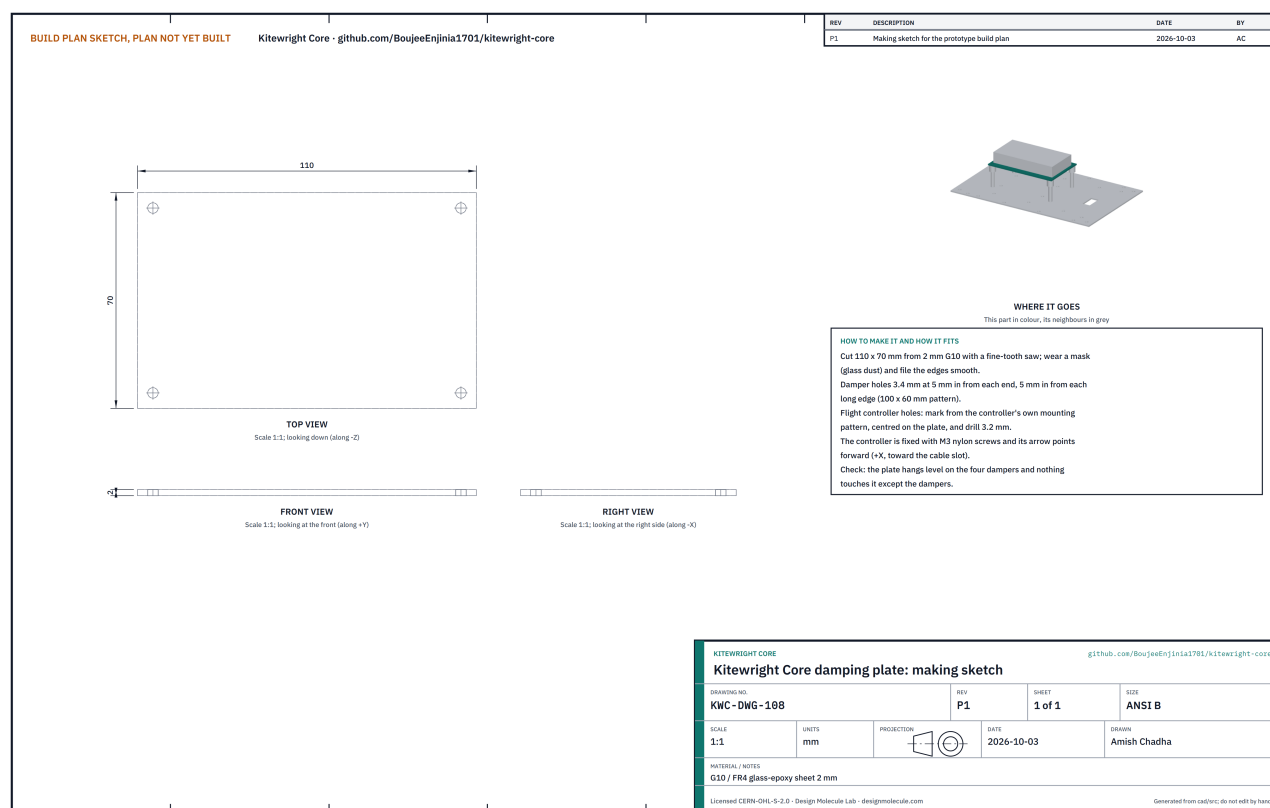


Figure 9. Making sketch KWC-DWG-108.

What it is and what it is made from. The plate the flight controller sits on: 110 x 70 mm of 2 mm G10 glass-epoxy sheet.

How to make it.

1. Cut 110 x 70 mm with a fine-tooth saw, wet or under extraction, and file the edges smooth.
2. Drill four 3.4 mm damper holes 5 mm in from each end and each long edge (a 100 x 60 mm pattern).
3. Mark the flight controller's own mounting pattern, centred, and drill 3.2 mm.

How it fits the parts next to it. It rests on the four silicone dampers, which sit on the tall standoffs; the controller is fixed to it with M3 nylon screws, its arrow pointing forward (Figure 17).

Check before moving on. It hangs level on the dampers and touches nothing else.

3.9 Bought components

Table 2. Bought components and what to do to them.

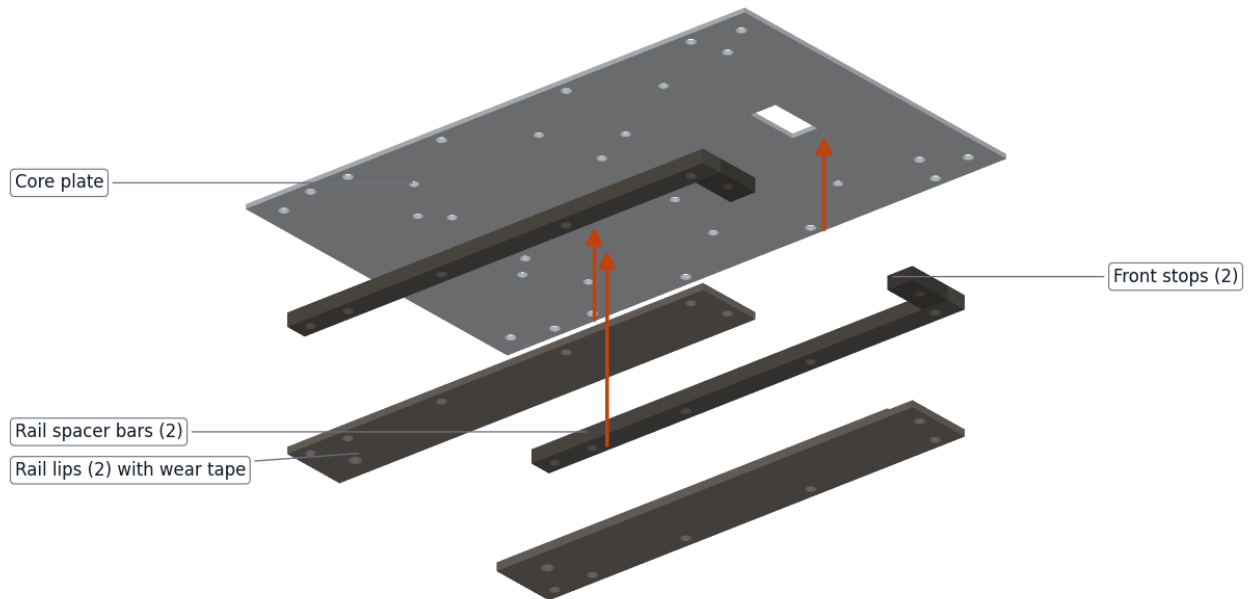
Component	What to buy (specification)	What to do to it
Corner spacers (4)	Aluminium round spacers 16 mm OD x 8 mm, M4 clearance	Nothing
Locking pins (2)	Stainless indexing plungers, M10 x 1, 5 mm hardened pin, at least 9 mm stroke, pull knob with a rest (lock-out) position, with jam nuts	Paint a red band round each pin just below the lip when the pin is withdrawn, so a withdrawn pin shows red
Clinch nuts (14)	Self-clinching M3 nuts for 2 mm aluminium, stainless	Press into the plate (3.1)
Standoffs (8)	M3 male-female aluminium: four 25 mm, four 8 mm	Nothing
Dampers (4)	Silicone grommet dampers for M3, about 10 mm tall, rated -40 °C	Choose hardness for the controller's mass
Flight controller	Pixhawk FMUV6X class with standard baseboard	Load released PX4 and the Kitewright parameter set
GNSS receiver and mast	Multi-band GNSS with compass; 10 mm OD carbon tube, 150 mm; M3 thumbscrew	Fix the receiver to the tube's top with its own mount
Power distribution board	18 to 60 V, 200 A peak, two inputs, two outputs, Hall current sensor, about 80 x 50 mm	Nothing
Power modules	Digital power monitor with 5.3 V supply, backup 5.3 V supply, payload switch (60 V, 8 A)	Nothing
Leads and connectors	8 AWG silicone wire, four AS150 halves	Cut four lead pairs about 100 mm long
Telemetry radio, receiver, antennas	900 MHz class radio pair, long-range control link receiver, three antennas and SMA bulkhead pigtails	Set the band and power for the country of use
DS-014 pigtail	40-pin connector pair per DS-014, 0.3 m pigtail, cable clamp	Wire to the controller and payload switch per the pin table
Safety switch, grommets, harness, gasket sheet, fasteners	As in bom/bom.csv	Nothing

■ 4. Putting it together

Step 1: rails onto the core plate

Step 1: rails onto the core plate (from below)

Spacer bars, front stops, then taped lips; ten M4 countersunk screws up through all three, nuts on top



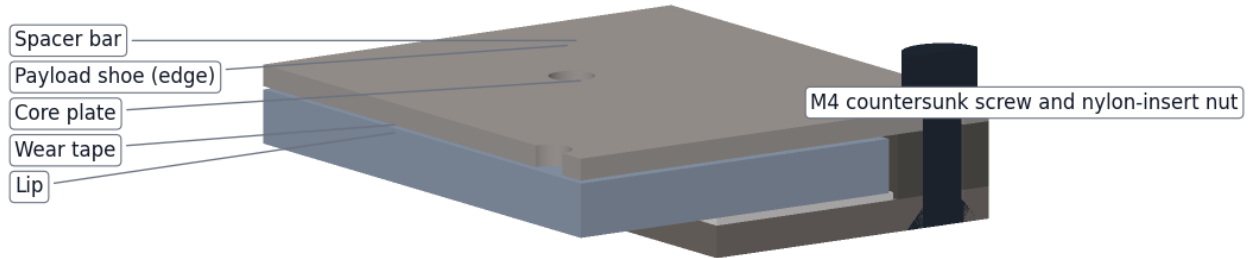
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github.com/BoujeeEnjinia1701/kitewright-core

Turn the plate upside down. Lay the two spacer bars along its long edges, the front stops at the front against their inner edges, then the taped lips, outer edges flush. Push ten M4 x 16 countersunk screws up through lip, spacer bar and plate (and the two stop screws through lip, stop and plate), and fit nylon-insert nuts on top. Use thread-locker; tighten to about 2 N·m. **Hold point:** do the shoe-offcut and feeler check of 3.4 now.

Joint 1: the plain rail, cut across at a screw

Shoe edge runs between plate and taped lip: 0.3 mm gap above, 1 mm at the side



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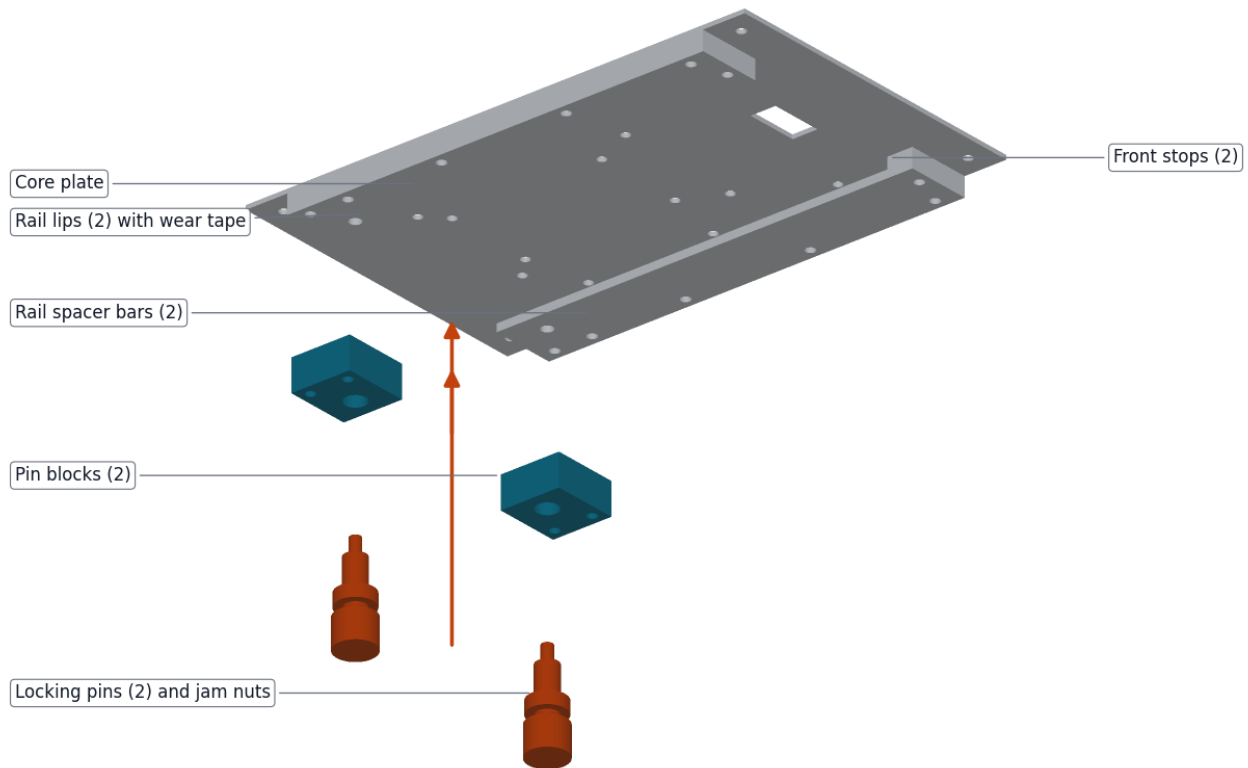
github.com/BoujeeEnjinia1701/kitewright-core

Figure 11. The rail cut across at a screw: the shoe's edge runs on the taped lip, 0.3 mm under the plate.

Step 2: pin blocks and locking pins

Step 2: pin blocks and locking pins

Blocks under the lips at the rear, M4 x 35 screws; screw each plunger in until its pin is 0.5 mm short of the plate



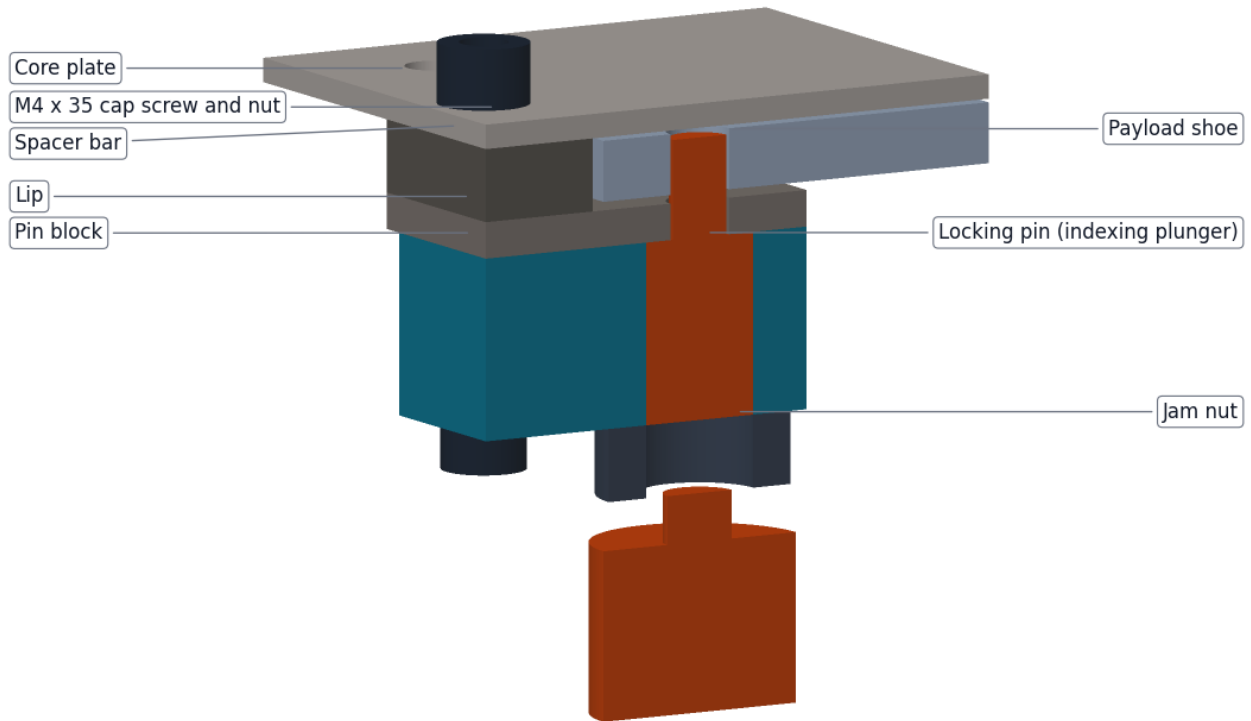
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Fit each pin block under its lip at the rear with two M4 x 35 cap screws up through block, lip, spacer bar and plate, nuts on top. Screw a plunger up into each block until, released, its pin tip is 0.5 mm short of the plate; lock it with the jam nut.

Joint 2: a locking pin, cut through its axis

Pin passes up through the lip into the shoe; pull the knob down and twist to hold it out



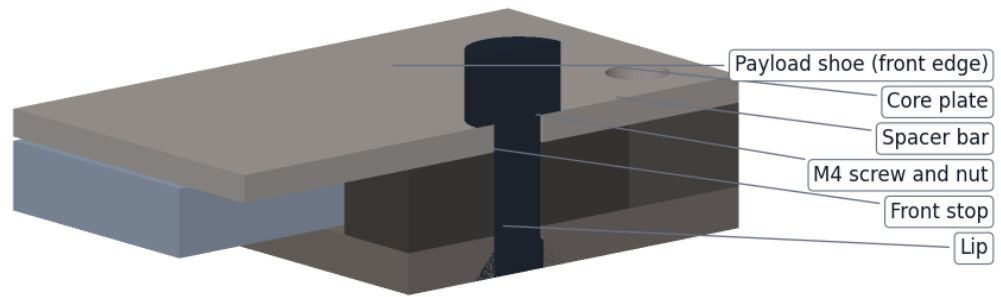
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Figure 13. A locking pin cut through its axis: the pin rises through the lip into the shoe.

Joint 3: a front stop, cut through its screw

The shoe's front edge stops against it; both pins then line up with their holes



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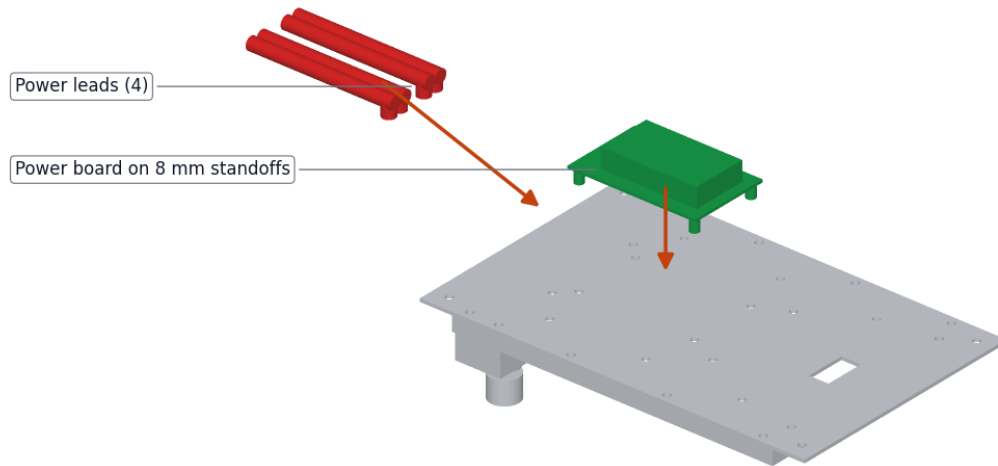
github.com/BoujeeEnjinia1701/kitewright-core

Figure 14. A front stop, clamped between plate and lip by the stop screw.

Step 3: power board and its leads

Step 3: power board and its leads

8 mm standoffs into the clinch nuts; solder the four leads to the board pads (no plugs yet)



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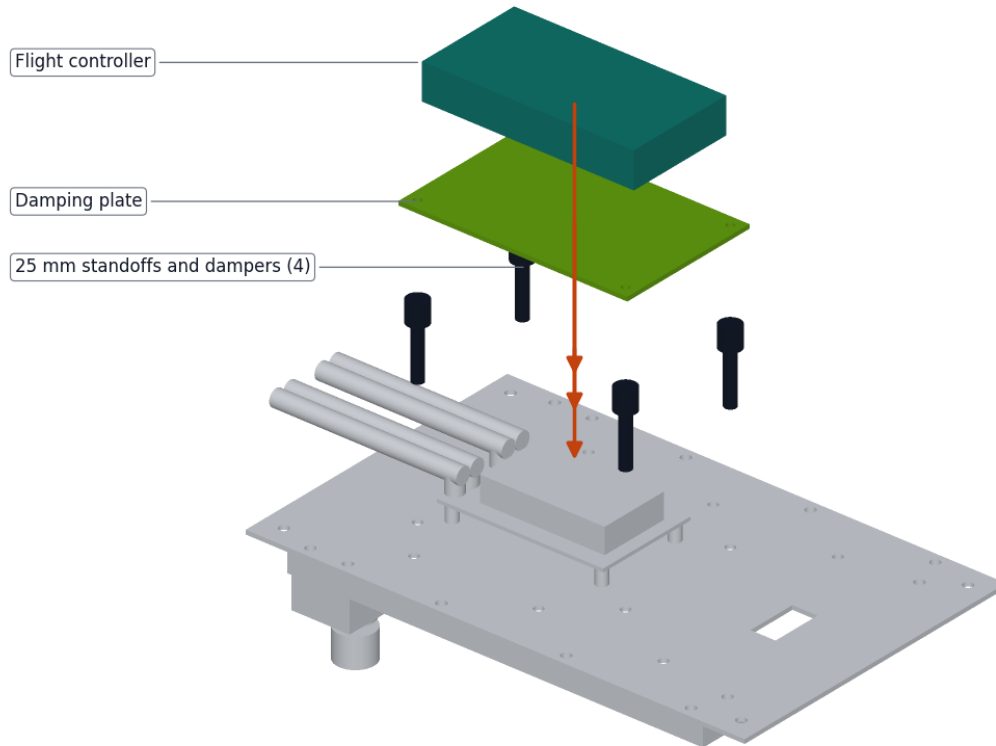
github.com/BoujeeEnjinia1701/kitewright-core

Turn the plate the right way up. Screw the four 8 mm standoffs into their clinch nuts and fix the power board on them with M3 screws. Solder the four lead pairs (two pack inputs, two frame outputs) to the board's pads, with no plugs on the far ends yet.

Step 4: flight controller on its dampers

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25 mm standoffs, dampers, damping plate; controller on M3 nylon screws, arrow forward



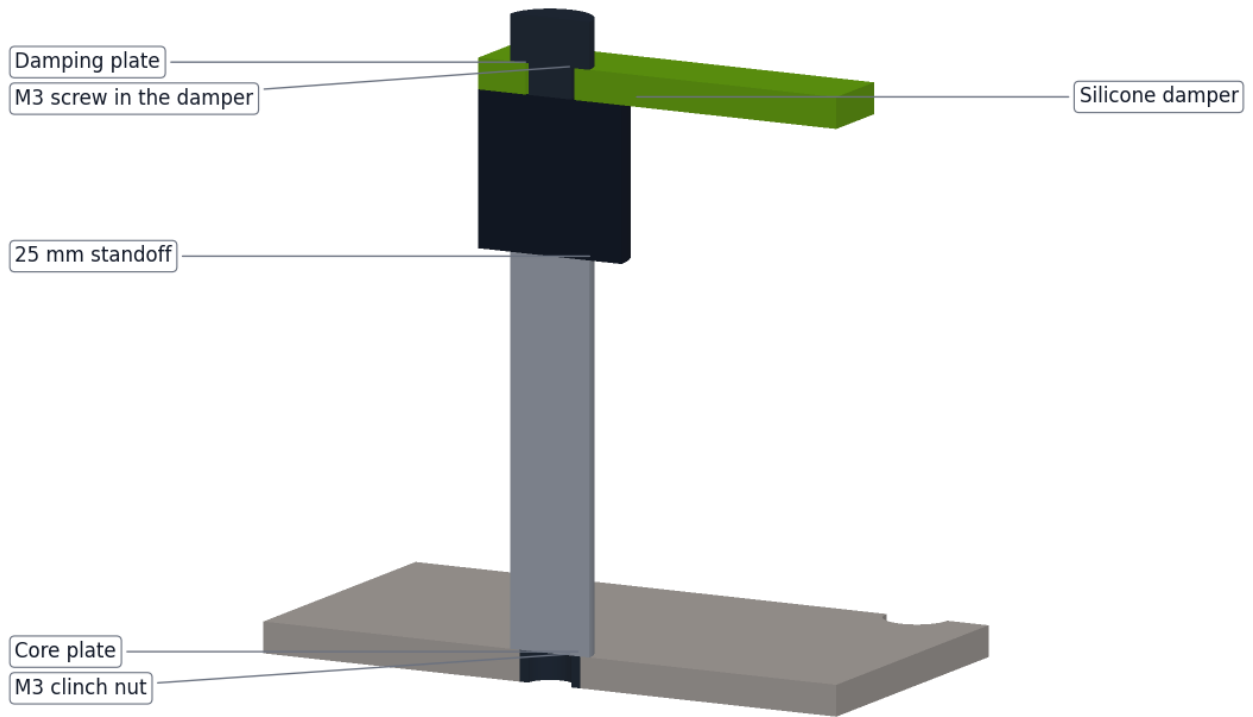
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Screw the four 25 mm standoffs into their clinch nuts, fit a damper on each, lay the damping plate on the dampers and fix it with M3 screws into the dampers. Fix the flight controller to the plate with M3 nylon screws, arrow forward.

Joint 5: flight controller damping, cut through a damper

Standoff screws into the flush clinch nut; the plate floats on four silicone dampers



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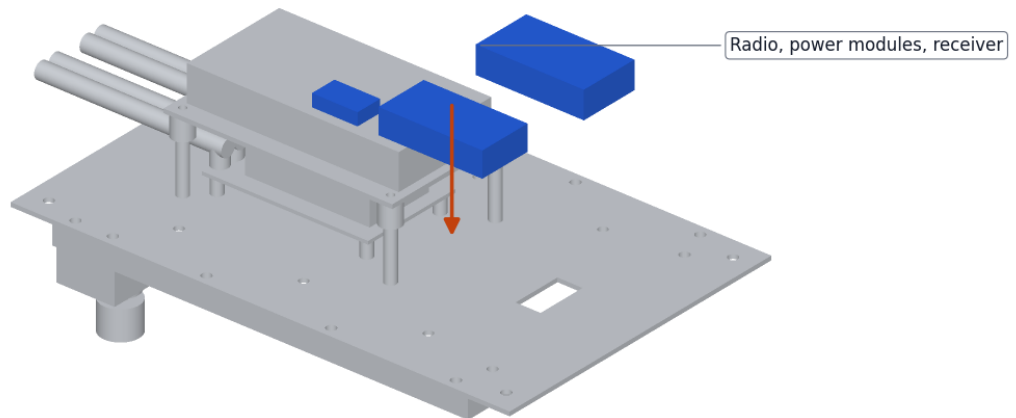
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Figure 17. Flight controller damping, cut through a damper.

Step 5: radio, power modules and receiver

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Foam-taped to the plate in front of the controller; plug the harness



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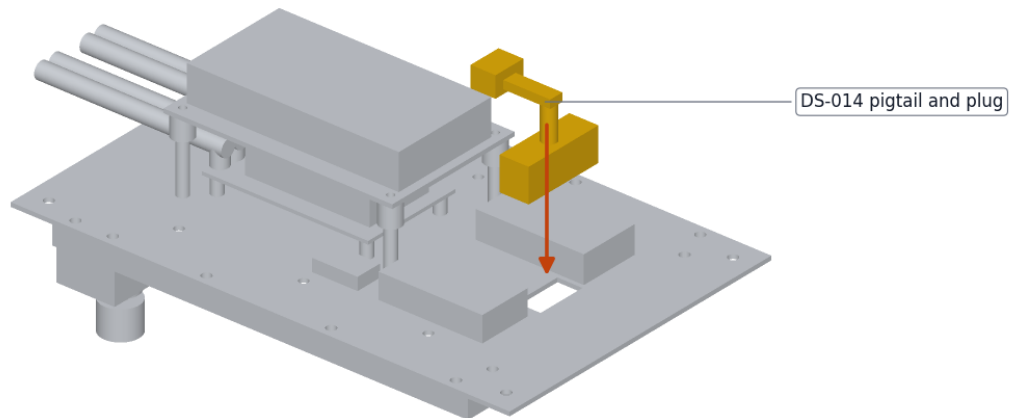
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Stick the telemetry radio (left front), power modules (right front) and control link receiver (right, under the damping plate) to the plate with foam tape, at room temperature. Plug in the harness: power modules to the controller's two power inputs, radio to a telemetry port, receiver to the control input, power board current sensor to its input, payload switch enable to a spare output.

Step 6: DS-014 pigtail

Step 6: DS-014 pigtail

Cable down through the slot, clamped to the plate; plug left hanging below



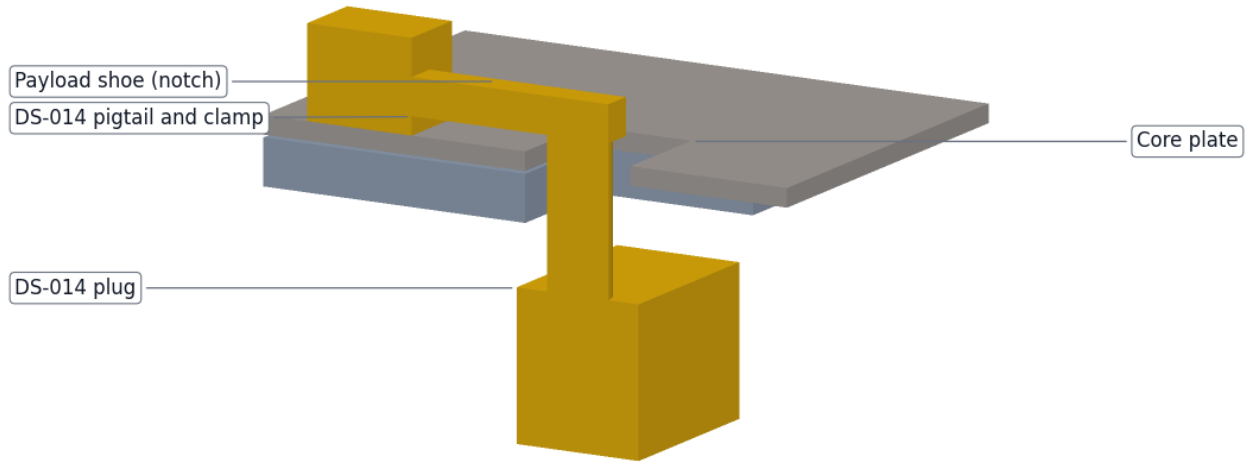
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Pass the pigtail's plug down through the cable slot from above and clamp the cable to the plate just behind the slot. Wire its power pins to the payload switch and its data pins to the controller's Ethernet, CAN, serial and trigger ports, per the DS-014 pin table.

Joint 4: the DS-014 pigtail, cut on the centre line

Cable drops through the 12 x 24 mm slot and the shoe's notch; plug hangs in front of the payload



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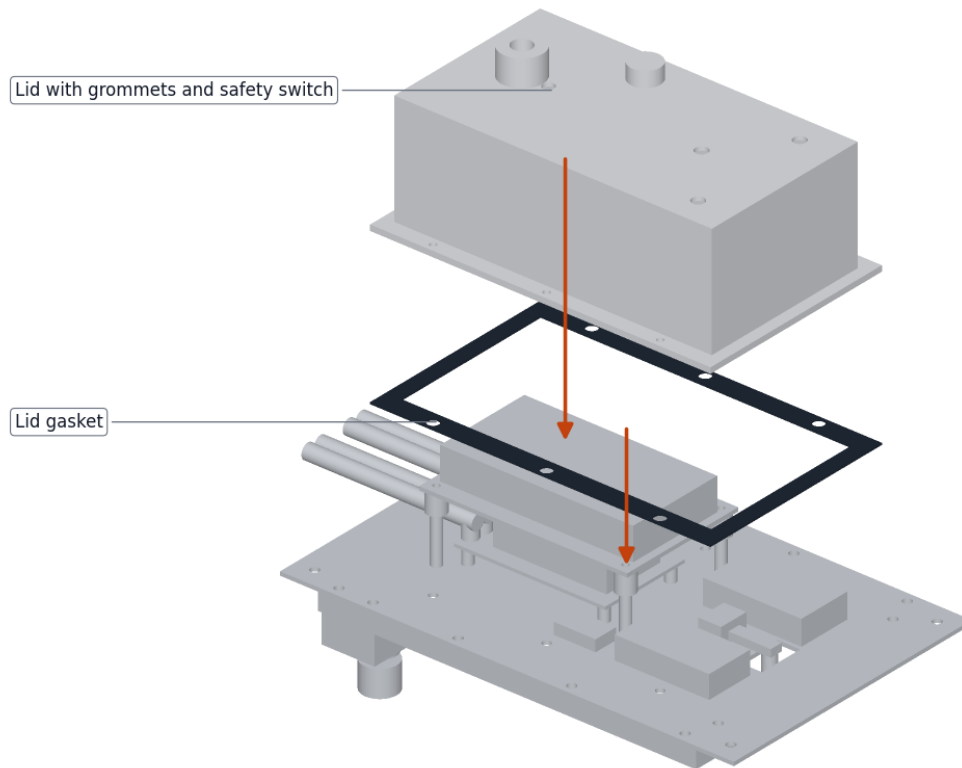
github.com/BoujeeEnjinia1701/kitewright-core

Figure 20. The pigtail through the slot and the shoe's notch; the plug hangs in front of the payload.

Step 7: gasket and lid

Step 7: gasket and lid

Thread the leads through the rear grommets, lower the lid, six M3 screws



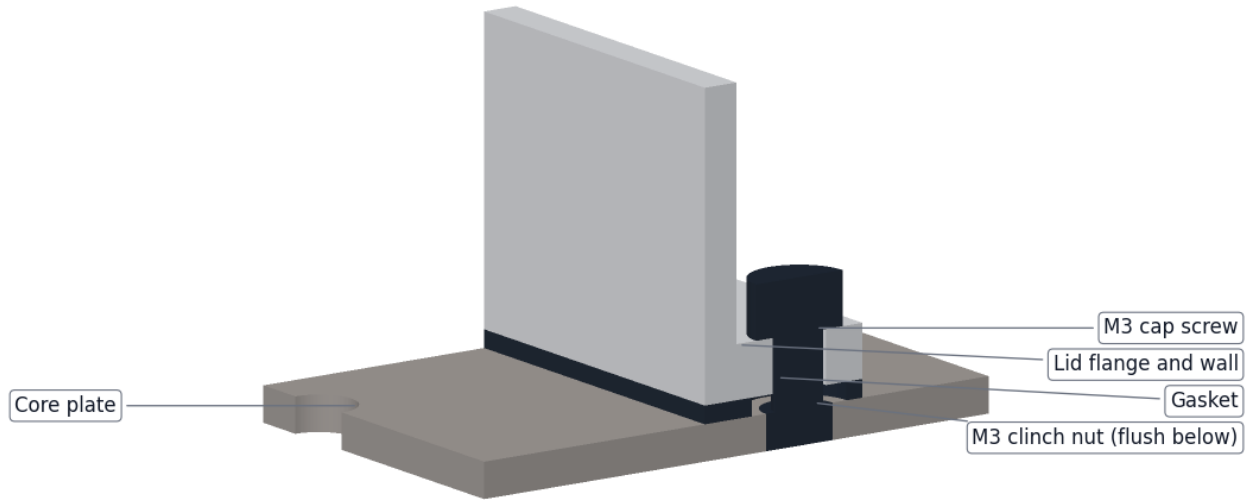
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Stick the gasket to the plate around the avionics. Fit the two grommets to the lid's rear wall and thread the four leads out through them, two leads per grommet. Lower the lid onto the gasket and fit six M3 x 8 cap screws through the flange into the clinch nuts, snug but not crushing the gasket.

Joint 6: lid flange to plate, cut through a screw

Six M3 screws into clinch nuts; the 1 mm foam gasket seals the flange



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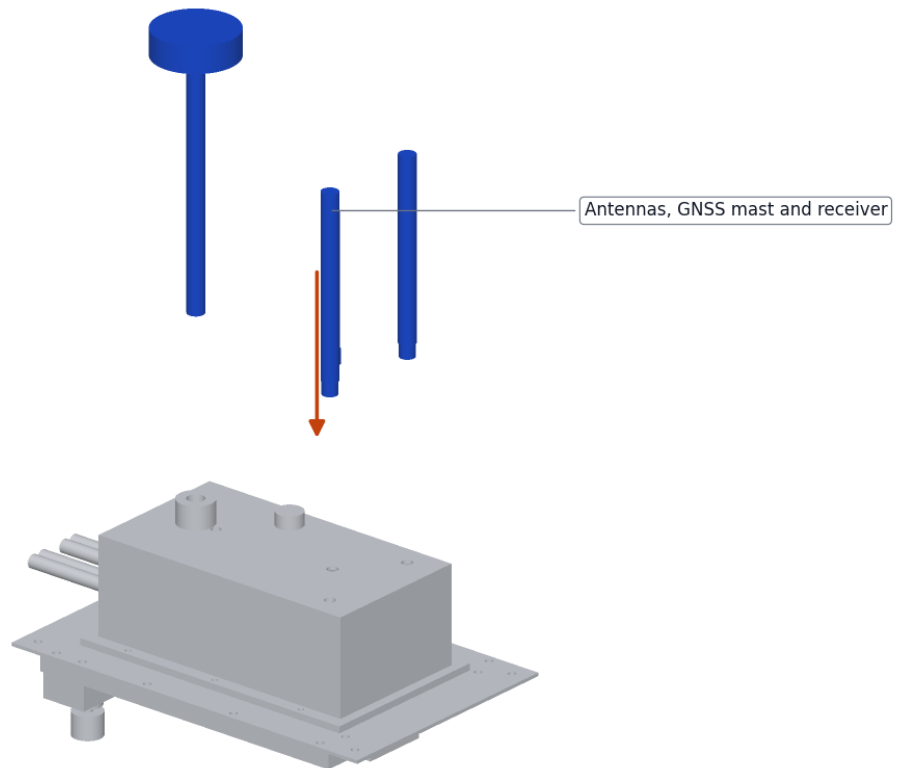
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Figure 22. Lid flange to plate: screw, gasket and the flush clinch nut.

Step 8: antennas, GNSS mast and receiver

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SMA bulkheads through the lid top; mast into its socket, thumbscrew



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

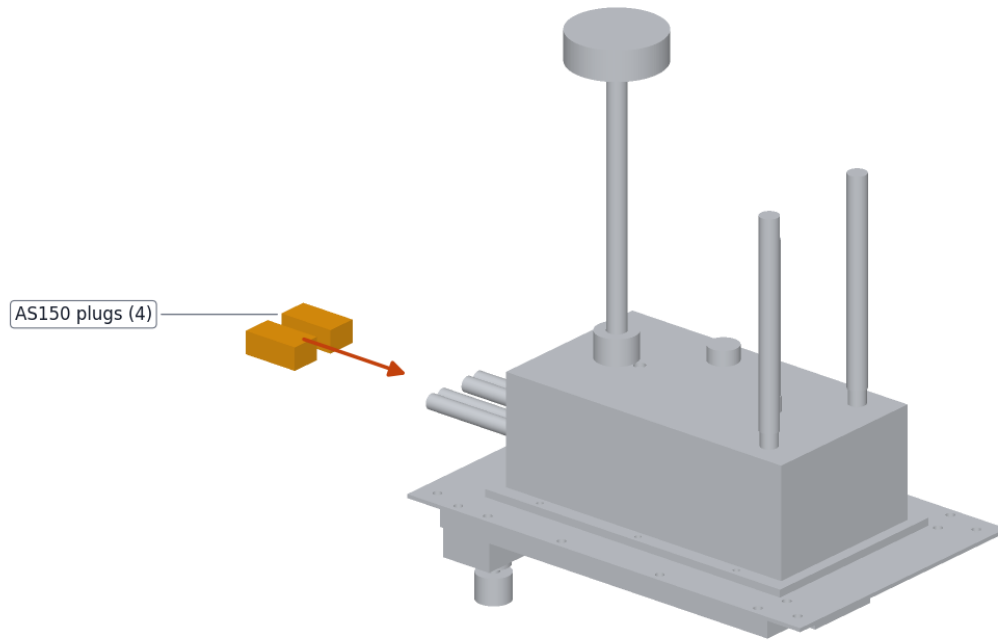
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Fit the three SMA bulkheads through the lid top and screw on the antennas. Press the safety switch into its hole. Push the mast into its socket, lock it with the thumbscrew, and run the GNSS cable in through the hole beside the boss.

Step 9: AS150 plugs on the leads

Step 9: AS150 plugs on the leads

Solder and heat-shrink; packs plug into the inputs, the frame into the outputs



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

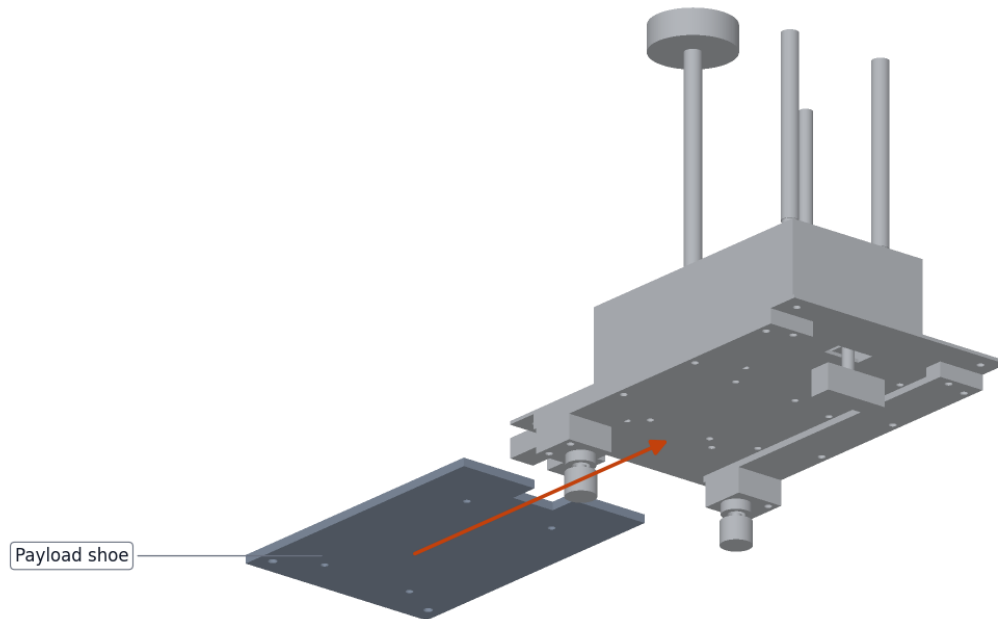
github.com/BoujeeEnjinia1701/kitewright-core

Solder an AS150 half to each lead pair, inputs and outputs of opposite gender so a pack cannot be plugged into an output, and heat-shrink them. Mark each lead "PACK 1", "PACK 2", "FRAME A", "FRAME B".

Step 10: payload shoe in from the rear (payload swap)

Step 10: payload shoe in from the rear (payload swap)

Pull and twist both pins out, slide to the front stops, twist back: both pins snap in, no red showing



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github.com/BoujeeEnjinia1701/kitewright-core

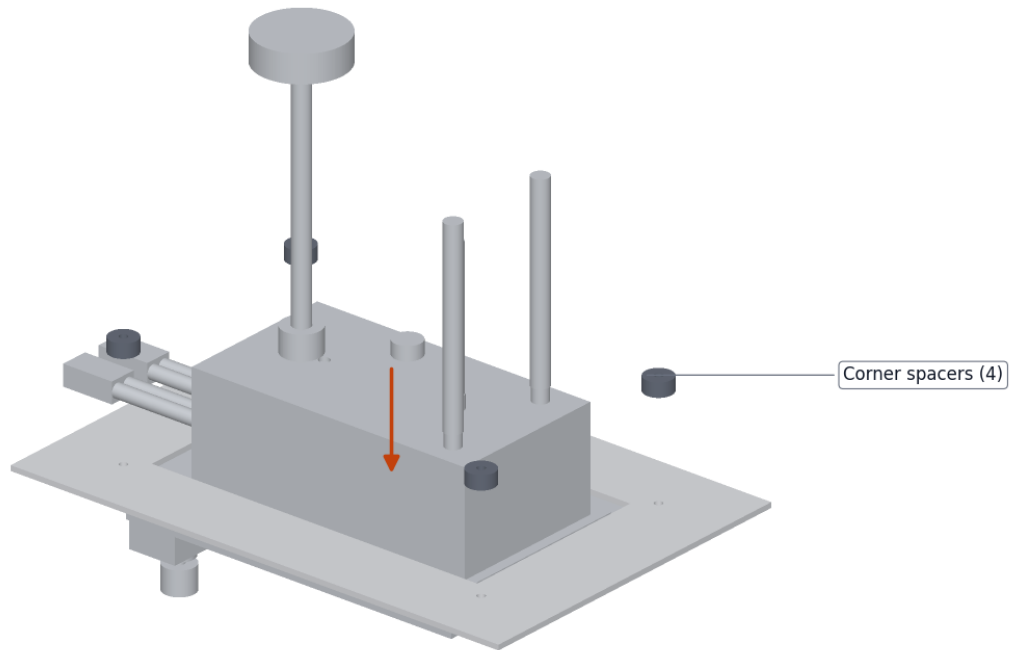
Pull both plunger knobs down and twist them to their rest. Slide the shoe into the rail from the rear until it meets the front stops. Twist both knobs back: the pins snap up into the shoe. Push the DS-014 plug into the payload's socket.

Hold point: no red band shows on either pin, and a firm tug does not move the shoe.

Step 11: into the frame

Step 11: into the frame

Corner spacers on the plate, lid up through the deck opening; four M4 bolts down through deck, spacers and plate



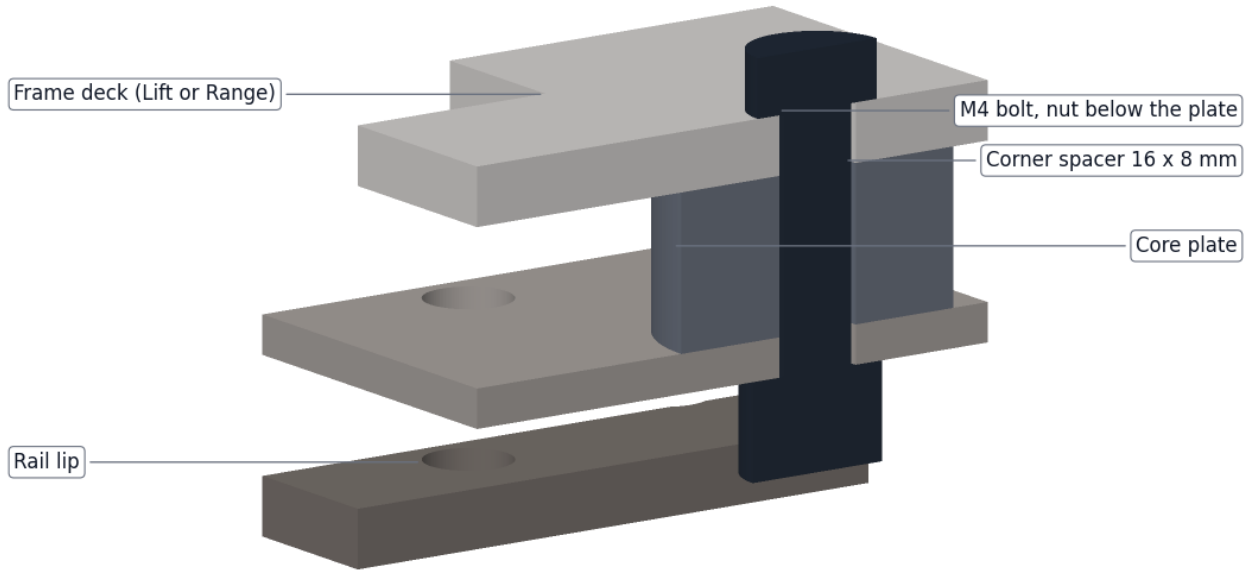
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/kitewright-core

Put a corner spacer on each frame hole of the plate, lift the core so the lid passes up through the frame deck's opening, and fit four M4 bolts down through deck, spacers and plate with nylon-insert nuts underneath. Plug the frame's speed controller leads into the two frame outputs.

Joint 7: core to frame deck at a corner, cut through the bolt

The core hangs under the deck on four M4 bolts; the lid rises through the deck opening



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Figure 27. Core to frame deck at a corner: the deck belongs to Kitewright Lift or Range.

■ 5. First checks

Table 3. First checks; a TRL 4 test report records the results.

Check	Requirement	How	Pass when
Rail fit	R2, R3	Slide the shoe the full rail length; feeler gauge above it	No binding; 0.2 to 0.4 mm gap; both pins drop in at the stops
Pin retention	R3	Pull each pin to its rest and back; tug the shoe with one pin withdrawn	Each pin alone stops the shoe; red band shows only when withdrawn
Proof load	R3	Hang 15 kg from the shoe for 1 min on the bench (core on a fixed stand, nobody beneath)	No permanent set, no slip
Payload swap	R2	Timed swap with winter gloves, three people, three times each	60 s or less, no tools
First power	R4, R5	Bench supply at 24 V, current limit 1 A, no packs	Controller boots; both 5.3 V rails in range; no part warm to the touch
Payload power	R4	Payload switch on into a 100 W load at 18 V and 50 V	Load holds 100 W; switch and pigtail stay below 60 °C
Cold boot	R5	Soak the core at -20 °C for 2 h, then power from the bench supply	Boots and logs normally
Interface	R1, R8	DS-014 continuity and a MAVLink payload heartbeat with a reference payload	Every pin in the table connected; payload recognised
Firmware	R10	Load the released PX4 build and the parameter set	No source changes; parameters load
Mass	R9	Weigh the core without the shoe and packs	Recorded against R9

■ 6. Safety stops

Work stops at each point below until what it says is true.

1. **Before first power.** Every solder joint inspected; no pack anywhere near the bench; the power board's inputs checked for shorts with a meter; the bench supply set to 24 V and 1 A.
2. **Before connecting a pack.** The bench checks of section 5 passed; inputs and outputs keyed by opposite-gender AS150 plugs and labelled; polarity checked at the plug with a meter; the pack warm enough to discharge (ColdCell reports at least 10 °C); a lithium fire plan, sand bucket and extinguisher at hand; the core on a non-flammable surface; nobody alone.
3. **Before loading the mount.** Both pins show no red band and the shoe passes the tug test; the core is held on a fixed stand, not in a frame; nobody under the load.
4. **Before fitting propellers or arming any frame.** That frame's own build plan safety stops apply; the core's safety switch works; failsafes for the altitude band are loaded.
5. **Before any flight with a payload.** Pin check and tug test done by a second person; the flight follows local rules and the site partner's permissions.

■ 7. Tools, skills and workspace

- Hacksaw or bandsaw, files, deburring tool, square and scribe; pillar drill with 3.2, 3.4, 4.2 (or 4.22), 4.5, 5.5, 6 and 9.0 mm drills; 90° countersink; M10 x 1 tap and wrench.

- Bench vice with two flat steel plates for the clinch nuts.
- FDM printer that can print ASA (enclosure, heated bed); heat-set insert tip.
- Temperature-controlled soldering station of 100 W or more for 8 AWG; heat-shrink and heat gun; crimp tools for JST-GH.
- Multimeter, current-limited bench supply (at least 60 V, 5 A), feeler gauges, kitchen scale.
- Skills: careful marking and drilling to 0.2 mm, tapping, soldering heavy leads, and following the PX4 setup guide.
- A bench with good light and ventilation, and a non-flammable area with a lithium fire plan for any work with packs.

■ 8. Where the numbers come from

- `cad/src/model.py` : every size and position, and the 614 constructability checks.
- `cad/drawings/KWC-DWG-001` : general arrangement; `KWC-DWG-101` to `108` : the making sketches.
- `docs/04-calcs/01-sizing.md` and `sizing.py` (`KWC-CAL-001`): loads, swap time, power, heat, altitude, mass and cost.
- `bom/bom.csv` : parts, specifications and prices.
- `docs/decisions/0001-trl2-review-decisions.md` and `0002-design-for-construction.md`.